Valve Inventory

User Manual

A task-by-task walkthrough of the desktop application. For the command-line equivalent of anything here, see the Technical Manual's command reference, or README.md.

Overview

The window opens on four tabs, described below, sharing one database. Nothing you do in one tab is hidden from the others - move stock in the Valves tab and the Browse tab's counts update the next time you search it.

- **Valves** - search, edit reference data, add/take/move/delete stock.
- **Bases / Sockets** - the same idea, for the sockets themselves rather than the valves that plug into them.
- **Browse** - a parametric filter across every held type, for “what do I have that could work here” questions.
- **Repair Bench** - for “I've got this valve out of a set I'm fixing - what is it, and what have I got that could stand in for it?”

The Valves tab

Boxes sidebar

Click a box to filter the results to it; click “All boxes” to clear. Click a column heading (Box / Types / Qty) to sort, click again to reverse.

Search row

Text, Function, Base, and the numeric fields (Heater V, Pa W, Freq MHz), which accept comparisons like `>20`, `<7`, `>=250` as well as an exact value.

> **Note:** Searching a type by name also pulls in stock of anything cross-referenced as its equivalent - search ECF80 and PCF80 stock shows up too, in blue, labelled which type it's equivalent to. This is a hint toward substitutes, not a claim they're necessarily interchangeable in every circuit.

Advanced... opens a dialog covering every remaining field - manufacturer, condition, family, gm, mu, power out, confidence, and whether a datasheet is linked. It edits the same underlying filters as the quick row, so the two stay in sync.

Results table

Click a heading to sort. Double-click a row to open its datasheet. **Amber** rows have unconfirmed (inferred) parameters; **blue** rows are equivalents pulled in by a search.

Detail panel

The selected row's type record, editable in place. **Save** keeps it inferred; **Save + confirm** marks it confirmed and the row turns from amber to black - that transition is the progress bar for working through the collection.

Below the fields, **Similar types** lists other held types with the same broad function and every shared electrical rating within 50% - candidates for a substitute with modification, not verified equivalents. Heater mismatches are flagged, not filtered out, since a dropping resistor or a different supply can often cover that. Double-click a suggestion to look it up.

Open datasheet opens the local PDF if there is one, otherwise falls back to an online source. **RadioMuseum** and **Web search** run a site-scoped or general search for whatever's selected.

Toolbar

Add stock creates the type automatically if it's new, classifying it from its designation. **Take**, **Move**, and **Delete lot** act on the selected row.

The Bases / Sockets tab

Valve bases and sockets aren't valves, so they're tracked in their own table rather than mixed into the general sundry catch-all. Same pattern as the Valves tab: search by base type or box, Add / Take / Move / Delete lot.

The Browse tab

A faceted filter across all held types, closer to a shopping-site filter panel than a search box.

- **Category, Base, Family, Confidence, Variable-mu** - dropdowns that *cascade*: picking one narrows what the others still offer, so you never land on an empty combination. Category is a coarser bucket than the raw function text (Triode, Double triode, Tetrode, Pentode, Triode-pentode, Rectifier, and so on) - specifically so it's useful as a filter instead of nearly matching one type each.
- **Numeric ratings** (Heater V/A, Va max, Pa max, gm, mu, Power out, Freq max) - an operator (< = > <= >=) plus a value picked from what's actually present in the data.
- **Name contains** - narrows the list as you type, e.g. “3cx” or “PL”.

Click a heading to sort. **Double-click a type** for a popup showing its full reference record, datasheet/web-search buttons, and a box-by-box breakdown of exactly where and how many you hold.

The Repair Bench tab

The workflow for a valve pulled out of a set on the bench: type its designation (and, optionally, which circuit stage it came from - IF amp, audio output, rectifier, and so on), then **Identify**.

If it's already in your database

Its reference data loads straight into the form on the left. On the right, **In stock now** shows anything you already hold of that exact type or a listed equivalent, and **Possible substitutes** lists other held types with the same broad function and every shared rating within 50% - the same candidate logic as the Valves tab's Similar types, but scoped to what's actually in stock, and with a held-quantity count. Double-click a substitute to switch the whole bench over to it, if that turns out to be the more interesting question.

If it's new to you

Open datasheet, **RadioMuseum**, and **Web search** work immediately off the typed designation, before anything is saved. **Copy research prompt** puts a ready-to-paste prompt on the clipboard, scoped to just this one type (a faster, single-item cousin of Tools > Generate research prompt...) - paste it into Claude, and it comes back in the same block format Apply researched data... expects.

Add to database creates a bare reference record (classified from the designation, no stock attached) so there's somewhere to save findings as you gather them. **Save** / **Save + confirm** work exactly as in the Valves tab detail panel - and immediately refresh the substitute list on the right using whatever you just entered, so you can see straight away whether the parameters you found open up any new candidates from stock.

Filling in reference data

New types start out with only what the naming convention can infer - a real datasheet reading is what actually confirms them. Three ways to close that gap:

By hand

Edit the fields in the detail panel and Save + confirm, as above.

With Claude - electrical parameters

Tools > Generate research prompt... writes a prompt (for your highest-quantity unconfirmed types) to a text file. Paste it into any Claude chat, save the reply, then Tools > Apply researched data.... Only what Claude actually confirmed is applied - a hedged finding (“could not verify”, “plausible”) is kept as a lead rather than marked confirmed, so nothing gets overclaimed.

With Claude - datasheet files

Tools > Generate datasheet download prompt... writes a prompt aimed at an agent with file and web access (Claude Code, not a plain chat, since it needs to write files to your disk). It tries the built-in fetcher first, then searches further for whatever's still missing, and saves PDFs directly into the local archive.

Adding stock in bulk

For more than a few lots at once, skip the Add-stock dialog:

- **Tools > Create upload template...** writes a blank CSV with the right columns, ready to fill in.

- **Tools > Import upload CSV...** reads a filled-in CSV back in - one row per lot, new types classified automatically, existing types just get more stock.
- **Tools > Generate CSV-building prompt...** writes a prompt for any Claude chat that interviews you (or reads whatever spreadsheet, notes, or photos you describe) and hands back a ready-to-import CSV - useful when your existing records aren't already in this shape.

Backup, export, and sharing

Backup

The database itself isn't the backup - the text snapshot is. Refresh it before ending a session of changes:

```
python3 snapshot.py
```

That's what belongs in version control; it's what a restore rebuilds from.

Export a spreadsheet

File > Export spreadsheet... writes a plain .xlsx for anyone who just wants to look, not use the tool.

Hand the whole thing to someone else

File > Export archive and tools (.zip)... bundles the code, docs, and a fresh snapshot into one file, with an option to strip the third-party descriptive text first (see the Technical Manual's note on that). The recipient unzips it and follows QUICKSTART.md, which is included.